

CPU, RTL, and measured-board latency

Python integer kernel

123621.090 ms

MEASURED

Exact C++ native CPU kernel

1777.700 ms

MEASURED

Verilator-generated RTL simulation

2456.708 ms

MEASURED

Pure RTL XSim active-cycle cross-check

36.012900 ms

DERIVED

FPGA board active-cycle latency

36.012900 ms

DERIVED

UART-paced transaction diagnostic

187144.751 ms

MEASURED

Source: [benchmarks/accelerator_benefit/figures/01_cpu_vs_rtl_latency_source.csv](#)

Limitation: Exact C++ is the native CPU baseline; Verilator is separately labeled RTL simulation runtime.